

Java Pattern Questions - Answers

Solid Square of Stars

Output:

```
*****
*****
*****
*****
*****
```

Java Code:

```
public class Pattern1 {
    public static void main(String[] args) {
        int n = 5;
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

Right-Aligned Triangle

Output:

```
  *
 **
 ***
 ****
```

Java Code:

```
public class Pattern2 {
    public static void main(String[] args) {
        int n = 4;
        for (int i = 1; i <= n; i++) {
            for (int space = 1; space <= n - i; space++) {
                System.out.print(" ");
            }
            for (int star = 1; star <= i; star++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

Inverted Number Triangle

Output:

```
12345
1234
123
12
1
```

Java Code:

```

public class Pattern3 {
    public static void main(String[] args) {
        int n = 5;
        for (int i = n; i >= 1; i--) {
            for (int j = 1; j <= i; j++) {
                System.out.print(j);
            }
            System.out.println();
        }
    }
}

```

Increasing Number Triangle

Output:

```

1
1 2
1 2 3
1 2 3 4

```

Java Code:

```

public class Pattern4 {
    public static void main(String[] args) {
        int n = 4;
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(j + " ");
            }
            System.out.println();
        }
    }
}

```

Continuous Numbers in Rows

Output:

```

1
2 3
4 5 6
7 8 9 10

```

Java Code:

```

public class Pattern5 {
    public static void main(String[] args) {
        int n = 4;
        int num = 1;
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(num + " ");
                num++;
            }
            System.out.println();
        }
    }
}

```

Hollow Square

Output:

```

*****
*   *
*   *

```

Java Code:

```
public class Pattern6 {
    public static void main(String[] args) {
        int n = 4;
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= n; j++) {
                if (i == 1 || i == n || j == 1 || j == n) {
                    System.out.print("*");
                } else {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }
}
```

Butterfly Pattern

Output:

```
*      *
**     **
***    ***
*****
***    ***
**     **
*      *
```

Java Code:

```
public class Pattern7 {
    public static void main(String[] args) {
        int n = 4;

        // Upper half
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            for (int j = 1; j <= 2 * (n - i); j++) {
                System.out.print(" ");
            }
            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println();
        }

        // Lower half
        for (int i = n - 1; i >= 1; i--) {
            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            for (int j = 1; j <= 2 * (n - i); j++) {
                System.out.print(" ");
            }
            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

Pascal's Triangle

Output:

```

    1
   1 1
  1 2 1
 1 3 3 1
1 4 6 4 1

```

Java Code:

```

public class Pattern8 {
    public static void main(String[] args) {
        int n = 5;
        for (int line = 1; line <= n; line++) {
            int number = 1;
            for (int space = 1; space <= n - line; space++) {
                System.out.print(" ");
            }
            for (int i = 1; i <= line; i++) {
                System.out.print(number + " ");
                number = number * (line - i) / i;
            }
            System.out.println();
        }
    }
}

```

X-Shaped Pattern**Output:**

```

  *   *   *
 * * * * *
*   *   *

```

Java Code:

```

public class Pattern9 {
    public static void main(String[] args) {
        int n = 3;
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= 2 * n - 1; j++) {
                if (j == i || j == (2 * n - i) || i == n) {
                    System.out.print("*");
                } else {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }
}

```